

# Vipul Sagar

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https://vipul-sagar.vercel.app/ | Delhi, India

- **AI Tools:** ChatGPT, Claude, GitHub Copilot, OpenAI API, Groq
- **Languages:** Python, SQL, Java
- **Backend:** FastAPI
- **AI/ML:** Machine Learning, Deep Learning, NLP, LLMs, RAG, Agentic AI, Finetuning, Prompt Engineering
- **Frameworks:** LangChain, LangGraph, LangSmith, scikit-learn, TensorFlow, Keras
- **Databases:** Qdrant, FAISS, ChromaDB, Redis, PostgreSQL, MongoDB, Neo4j, MySQL
- **Analytics:** Excel, Power BI, Data Cleaning
- **Tools:** Git, Docker, Jupyter, VS Code, n8n, Supabase

## PROFESSIONAL EXPERIENCE

### Data Analytics Intern

Indian Cybercrime Coordination Center (I4C), MHA

Jun 2025 – Aug 2025

New Delhi, India

- Designed and developed interactive dashboards for cybercrime analysis, serving 1,000+ law enforcement officials.
- Built graph-based data models using **Neo4j** to visualize relationships between cases, suspects, and digital evidence.
- Improved data reporting efficiency by 100%, enabling real-time insights for critical decision-making.

## EDUCATION

### Degree / School

B.Tech – AI & Data Science, Vivekananda Institute of Professional Studies (GGSIPU)

### Year

2023–2027

### Score

9 CGPA

## PROJECTS

### ArXiv Research Agent – Agentic RAG System | github.com/vipulsagar25/skyclad-assignment-vipul

- Built an end-to-end agentic RAG system using LangGraph with an 8-node conditional workflow over 100+ recent cs.AI arXiv papers, enabling autonomous retrieval, clarification, web search, refusal routing, and context-aware query enrichment.
- Improved retrieval relevance using BGE-M3 hybrid search and reranking on Qdrant, achieving a 395% retrieval improvement and increasing evaluation scores from 9/20 to 17/20 in ablation testing through optimized chunking and advanced hybrid retrieval strategies.
- Integrated LangSmith observability for detailed per-node latency, token usage, and real-time agent workflow tracking.

### MovieGallery – Production-Grade Recommendation Engine | github.com/vipulsagar25/MovieGallery

- Built a full-stack movie recommendation system serving 33.8M ratings across 330K users and 86K movies using ALS collaborative filtering with FAISS vector search and scalable asynchronous recommendation pipelines for personalized recommendations.
- Architected a decoupled ML worker with 100 async coroutines, Redis-backed queues, ORJSON serialization, resilient retry handling, and background task processing, achieving sub-100ms API latency with zero failures across 134K+ HTTP requests during load testing.
- Achieved 968 RPS with sub-4ms minimum latency using a highly optimized FAISS, Redis, and FastAPI inference pipeline.

### Clinical Co-Pilot – RAG Healthcare Assistant | github.com/vipulsagar25/Clinical-Co-pilot

- Built end-to-end RAG pipeline using LLMs and **ChromaDB** with semantic retrieval for grounded clinical responses.
- Implemented document chunking, embeddings, and a **FastAPI**-based backend for high-concurrency inference.

## ACHIEVEMENTS & CERTIFICATIONS

- **Smart Delhi Ideathon 2025:** 3rd Runner Up among 500+ teams, recognized by Lt. Governor of Delhi.
- **Smart India Hackathon (SIH):** 2nd Runner Up – Internal University Round.
- **HackLLM – IIIT Delhi:** Top 10 Teams among 500+ teams.
- **AI for Quality Hackathon:** Top 10 position in Rapid Fire.
- **Job Simulations:** Accenture (Data Analytics), BCG X (Gen AI) | Excel Certification (Coursera).